Graphs in Machine Learning

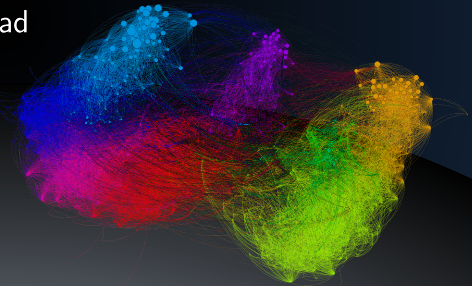
Small-World Phenomena

Erdős Numbers and Disease Spread

Michal Valko

Isara Labs

Partially based on material by: Andreas Krause, Branislav Kveton, Michael Kearns



Erdős number project

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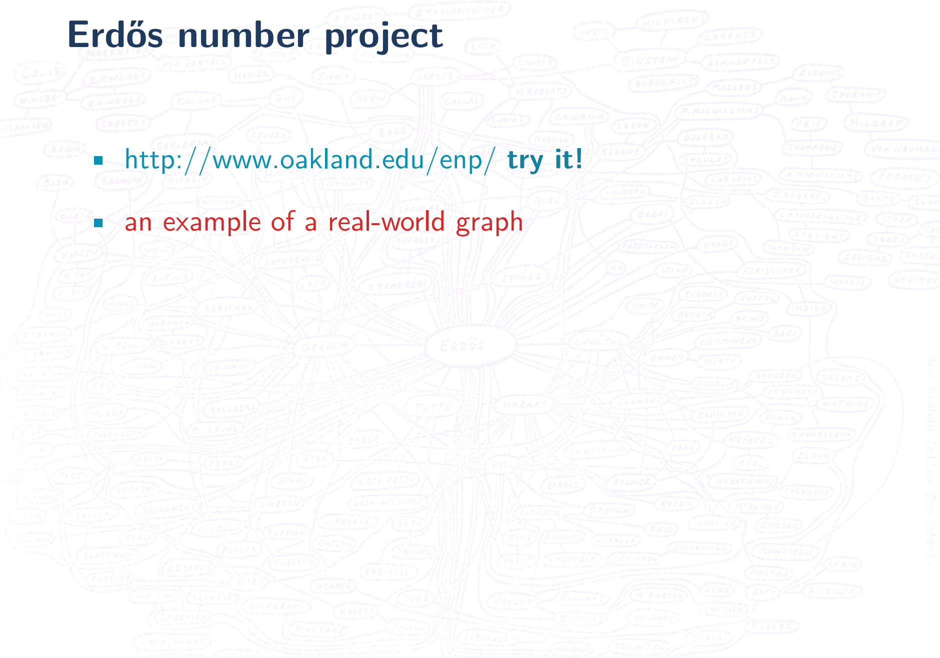

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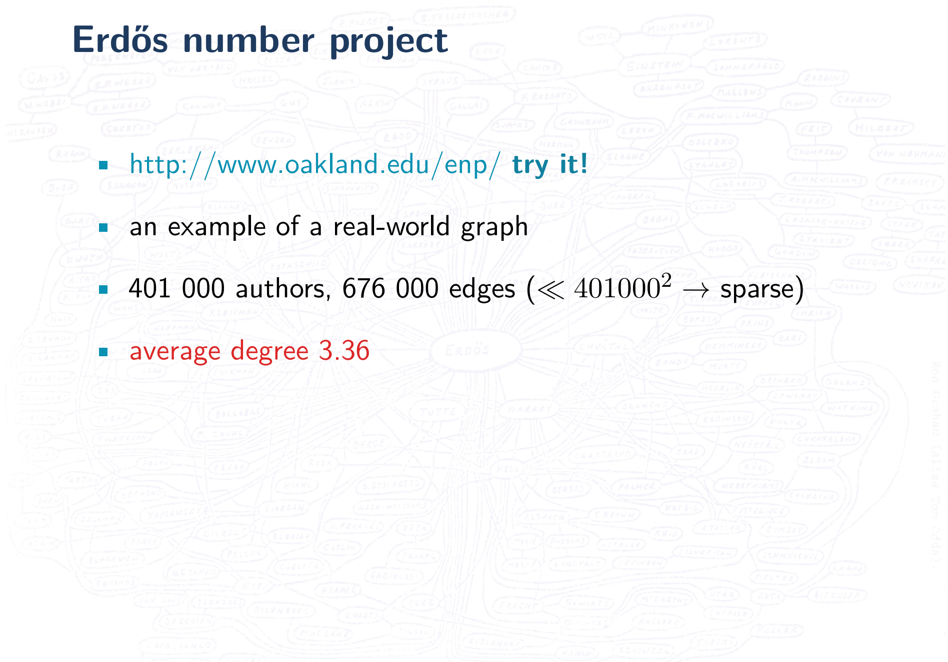

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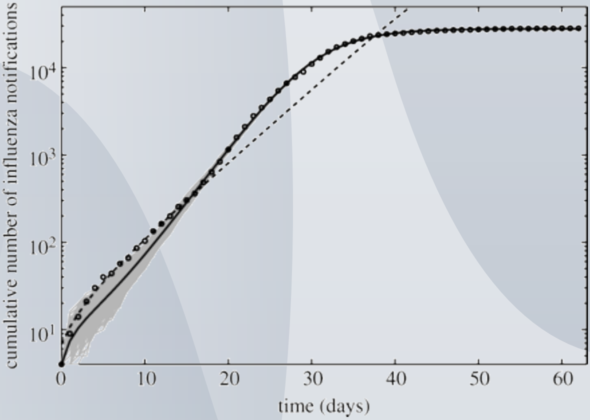
- heavy tail

Spanish flu in San Francisco 1918–1919

Small-world phenomenon and diseases



http://rsif.royalsocietypublishing.org/content/4/12/155

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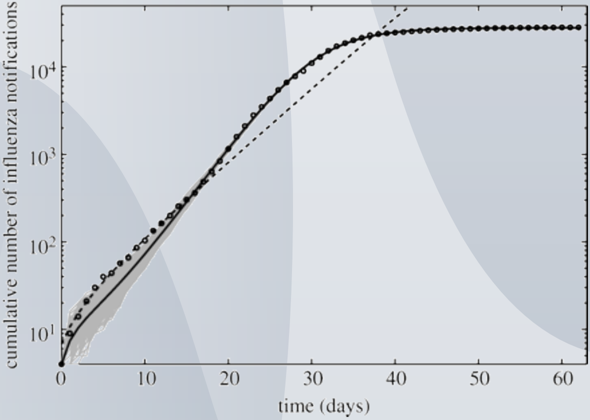

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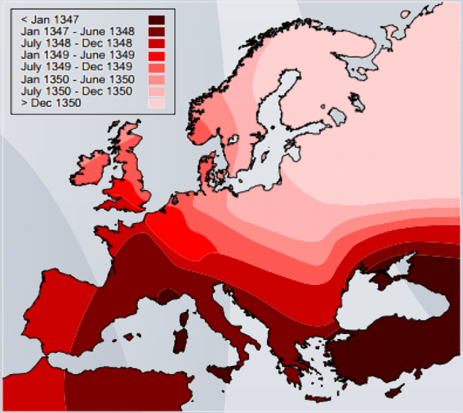
Small world: Obvious?

Black death!

Black death: spread



source: catholic.org

https://www.youtube.com/watch?v=EEK6c9Bh5CQ

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https://misovalko.github.io/mva-ml-graphs.html